Cogent Food & Agriculture

ISSN: 2331-1932 (Online) Journal homepage: www.tandfonline.com/journals/oafa20

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To cite this article: Fahmi Alif Ardiansyah, Ahmad Ridwan & Siti Azizah (2026) E-commerce adoption in livestock product marketing: a systematic review of challenges and opportunities in selected developing countries, Cogent Food & Agriculture, 12:1, 2703895, DOI: 10.1080/23311932.2026.2703895

To link to this article: https://doi.org/10.1080/23311932.2026.2703895

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COGENT FOOD & AGRICULTURE
2026, VOL. 12, NO. 1, 2703895
https://doi.org/10.1080/23311932.2026.2703895

FOOD SCIENCE & TECHNOLOGY | REVIEW ARTICLE

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E-commerce adoption in livestock product marketing: a systematic review of challenges and opportunities in selected developing countries

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ABSTRACT

E-commerce offers transformative potential for livestock product marketing in developing countries by reducing intermediary dependency, enhancing market access, and enabling price premiums through digital quality signalling. However, systematic evidence on adoption dynamics remains limited. This study presents a systematic review guided by PRISMA 2020, drawing on Scopus and Web of Science (2016–2026). Following rigorous screening, 11 peer-reviewed articles were synthesised using inductive thematic analysis, reflecting the nascent state of the field. Five thematic clusters emerged: e-commerce adoption and behavioural factors; consumer behaviour and market access; digital surveillance and food safety regulation; value chain resilience and digital pivoting; and ICT integration and supply chain performance. Adoption remains critically low, constrained by inadequate digital infrastructure, low digital literacy, capital limitations, and weak regulatory frameworks, though substantial opportunities exist through direct market access, traceability-based premiums, and ICT-enabled efficiency. Agricultural economics frameworks dominate theoretically, mainstream technology adoption models remain absent, and studies are geographically concentrated in China, limiting generalisability. Critical gaps include limited longitudinal designs, smallholder perspectives, and gender-inclusive analyses.

ARTICLE HISTORY
Received 13 March 2026
Revised 23 June 2026
Accepted 10 July 2026

KEYWORDS
E-commerce adoption; livestock product marketing; developing countries; smallholder farmers; systematic review

SUBJECTS
Internet / Digital Marketing / e-Marketing; Agricultural Development; Agricultural Economics

1. Introduction

The livestock sector constitutes one of the most critical components of the global food security system, particularly in developing countries where livestock commodities such as meat, dairy, and eggs serve as the primary sources of animal protein for populations with limited purchasing power (FAO, 2023). Globally, more than one billion people depend wholly or partially on the livestock sector for their livelihoods, with the largest proportion residing in South Asia, Southeast Asia, Sub-Saharan Africa, and Latin America (World Bank, 2022). In developing countries, this sector is predominantly characterised by small-scale livestock enterprises confronting numerous structural constraints: limited market access, inadequate logistical infrastructure, severe price information asymmetries, and substantial dependence on intermediaries (middlemen) that diminish the value share received by farmers at the upstream level of the supply chain (Zeng et al., 2017). Evidence from Sub-Saharan Africa further illustrates that smallholder cattle business systems are shaped by geographical market opportunities, market access constraints, intermediary dependence, and sustainability challenges, indicating that digital livestock marketing interventions should be embedded within wider livestock business-system realities rather than treated as purely technological solutions (Siankwilimba et al., 2024).

A systematic literature review in the field of agri-food e-commerce by Zhang and Berghäll (2021) using the Service-Dominant Logic framework identified that agricultural e-commerce research is still

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Supplemental data for this article can be accessed online at https://doi.org/10.1080/23311932.2026.2703895.

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dominated by a platform perspective and has not sufficiently explored the value dimensions for small-scale producers. The emergence and rapid development of e-commerce over the past decade has inaugurated a new paradigm in the governance of livestock product marketing. E-commerce is defined as the activity of buying and selling products and services facilitated by digital information and communication technology networks, encompassing various business models ranging from Business-to-Consumer (B2C), Business-to-Business (B2B), and Consumer-to-Consumer (C2C), to Social Commerce based on social media platforms (Laudon & Traver, 2021). For the livestock sector in developing countries, e-commerce theoretically offers transformative solutions to overcome traditional market barriers: the elimination of inefficient intermediaries, geographic expansion of market reach, enhanced price transparency, and the opening of access to premium consumer segments willing to pay higher prices for quality and certified livestock products (Qiu et al., 2025; Yu et al., 2023).

Nevertheless, the realisation of this potential confronts a range of complex barriers that are technical, institutional, and socio-cultural in nature. Empirical literature indicates that the rate of e-commerce adoption among smallholder farmers in developing countries remains extremely low, with key barriers including limited digital infrastructure (internet connectivity and cold chain), low digital literacy, limited business capital, distrust of online transactions, the absence of adequate regulatory frameworks, and the perishable characteristics of livestock products that complicate the management of long-distance logistics (Choruma et al., 2024; Tian & Wu, 2024; Zeng et al., 2017). The complexity of these barriers indicates that effective e-commerce adoption strategies in the livestock sector cannot be formulated generically, but instead require an in-depth understanding of the specific dynamics within the developing country context.

Although the literature on agricultural e-commerce generally has expanded rapidly in recent years as evidenced by the bibliometric analysis of Jusoh et al. (2025) identifying 1,342 Scopus-indexed documents on this topic from 2016 to 2026 systematic reviews that specifically focus on e-commerce adoption in livestock product marketing in developing countries remain highly limited. The majority of existing literature reviews have either broader coverage (agriculture in general) or narrower scope (a single country or specific commodity), and are consequently unable to produce a comprehensive, cross-contextual picture of the specific challenges and opportunities facing the livestock sector (Choruma et al., 2024; Zeng et al., 2017).

This literature gap carries serious policy implications. Without a systematic and evidence-based understanding of the factors that determine the success and failure of e-commerce adoption in the livestock sector of developing countries, policymakers, industry practitioners, and researchers lack an adequate foundation upon which to design effective and contextual interventions. Therefore, a comprehensive systematic review in this area is urgently required to map the current state of knowledge, identify consistent patterns, and direct future research and policy agendas.

1.1. Research questions

The research questions in this review were formulated using the Population-Interest-Context (PICo) framework, which is commonly employed in systematic reviews in the social sciences and management disciplines (Mohamad Shaffril et al., 2020). In this study, population refers to actors in the livestock sector in developing countries (farmers, SMEs, and agribusiness companies); interest pertains to e-commerce adoption in livestock product marketing activities; and context encompasses the conditions of developing countries as classified by the World Bank. Based on this framework, three primary research questions were formulated as follows:

- RQ1: What are the challenges and opportunities associated with e-commerce adoption in livestock product marketing in developing countries?
- RQ2: What theoretical frameworks have been employed to examine e-commerce adoption in livestock product marketing in developing countries?
- RQ3: What are the existing literature gaps and what future research directions are recommended to advance e-commerce adoption in the livestock agribusiness sector of developing countries?

Through this review, researchers, practitioners, and policymakers are expected to gain a structured and evidence-based understanding of the current state of e-commerce adoption in the livestock sector of developing countries, while also obtaining clear guidance on the most necessary future research directions and policy interventions to promote sustainable and inclusive digital transformation of the livestock sector.

2. Methodology

2.1. Systematic review guide

This systematic review was conducted following the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) 2020 guidelines developed by Page et al. (2021). PRISMA 2020 represents an update to the PRISMA 2009 guidelines and is currently the most widely adopted standard for systematic review reporting in the international scientific community, having been cited by more than 60,000 research reports indexed in Scopus (Page et al., 2021). The selection of PRISMA 2020 as the methodological framework is based on its capacity to guide researchers through the process of identification, selection, eligibility assessment, and synthesis of studies in a systematic and transparent manner, as well as its effectiveness in minimising bias in the literature selection process (Page et al., 2021; Tranfield et al., 2003).

The systematic review process encompassed six primary stages: (1) formulation of research questions using the PICo framework; (2) identification of literature through systematic searches of relevant databases; (3) screening based on inclusion and exclusion criteria; (4) eligibility assessment through full-text reading; (5) methodological quality appraisal; and (6) data extraction and thematic synthesis. Each stage was transparently documented in the PRISMA 2020 flow diagram to enable independent replication and verification of the review process.

2.2. Literature search strategy

2.2.1. Identification

Relevant literature was identified through systematic searches of two primary databases: Scopus and Web of Science (WoS). Both databases were selected because they are recognised as the most comprehensive sources of peer-reviewed academic literature in the fields of management science, agricultural economics, and e-commerce studies (Falagas et al., 2008; Zhu & Liu, 2020). Scopus served as the primary database given its broader coverage of multidisciplinary literature and availability of advanced field-code-based search features that facilitate structured searching (Mohamad Shaffril et al., 2020). Web of Science was used as a secondary database to ensure completeness of coverage, particularly for literature in agricultural economics and agribusiness.

Search strings were developed based on three primary conceptual blocks consistent with the PICo framework: (1) e-commerce and digital marketing terminology; (2) livestock sector and animal product terminology; and (3) developing country and adoption terminology. Boolean operators (AND, OR), wildcards (*), and phrase searches ('…') were employed to optimise the comprehensiveness and precision of search results. The final search strings used are presented in Table 1. The database search was conducted on 19 June 2026.

The revised search strings were developed to explicitly incorporate core e-commerce terminology alongside social commerce and digital marketing terms, addressing the recognised limitation of the original search strategy. The updated strings now include terms such as 'e-commerce,' 'electronic commerce,' 'online marketplace,' 'online retail,' 'digital marketplace,' 'digital trade,' and 'social commerce,' combined with livestock product terminology using clearly structured Boolean parentheses (AND/OR). This revision ensures that both formal e-commerce platform-based trade and informal social commerce practices relevant to livestock product marketing in developing countries are captured. It is nonetheless acknowledged that digital trade in smallholder livestock contexts frequently occurs through social media platforms such as Facebook, WhatsApp, and Instagram, and that restricting the search exclusively to formal

Table 1. Boolean search strings in Scopus and Web of Science databases.

Source	Keyword string
Scopus	TITLE-ABS-KEY (('e-commerce' OR 'ecommerce' OR 'electronic commerce' OR 'online marketplace*' OR 'digital marketplace*' OR 'online selling' OR 'online retail*' OR 'digital trade' OR 'social commerce' OR 'digital marketing' OR 'online marketing' OR 'social media marketing' OR 'social media advertising' OR 'influencer marketing') AND ('purchase intention' OR 'buying intention' OR 'consumer behavior' OR 'consumer behaviour' OR 'purchase decision' OR 'buying behavior' OR 'adoption' OR 'technology adoption') AND ('livestock' OR 'meat' OR 'beef' OR 'honey' OR 'poultry' OR 'chicken' OR 'dairy' OR 'milk' OR 'egg' OR 'mutton' OR 'goat meat' OR 'animal product*' OR 'livestock product*'))
Web of Science	TS = (('e-commerce' OR 'ecommerce' OR 'electronic commerce' OR 'online marketplace*' OR 'digital marketplace*' OR 'online selling' OR 'online retail*' OR 'digital trade' OR 'social commerce' OR 'digital marketing' OR 'online marketing' OR 'social media marketing' OR 'social media advertising' OR 'influencer marketing') AND ('purchase intention' OR 'buying intention' OR 'consumer behavior' OR 'consumer behaviour' OR 'purchase decision' OR 'buying behavior' OR 'adoption' OR 'technology adoption') AND ('livestock' OR 'meat' OR 'beef' OR 'honey' OR 'poultry' OR 'chicken' OR 'dairy' OR 'milk' OR 'egg' OR 'mutton' OR 'goat meat' OR 'animal product*' OR 'livestock product*'))

Source: Authors' own work.

Table 2. Inclusion and exclusion criteria.

Criteria	Inclusion	Exclusion
Topic	Studies specifically addressing e-commerce adoption in livestock product digital marketing in developing countries	Studies unrelated to e-commerce, livestock products, or developing country contexts
Population	Smallholder farmers, agribusiness actors, cooperatives, consumers, and policymakers involved in livestock product marketing	Studies focusing exclusively on developed countries without comparison to developing countries
Intervention	E-commerce platforms, digital marketplaces, online marketing, social commerce, and digital agri-food platforms	Studies focusing solely on conventional/traditional marketing without any digital component
Outcome	Studies reporting challenges, opportunities, barriers, benefits, theoretical frameworks, or adoption factors related to e-commerce in livestock marketing	Studies with no measurable or identifiable outcomes related to e-commerce adoption
Publication Type	Peer-reviewed journal articles indexed in Scopus & WOS	Theses, dissertations, book chapters, grey literature, editorials, and non-peer-reviewed publications
Language	English	Non-English publications
Year	2016–2026	Publications before 2016
Accessibility	Full-text articles available for retrieval	Articles available in abstract only or behind restricted access
Study Design	Quantitative, qualitative, and mixed-method studies	Opinion pieces, commentaries, and letters to the editor

Source: Authors' own work.

e-commerce terminology would risk excluding a substantial body of relevant literature on these practices. Future reviews are encouraged to further combine social commerce terminology with digital marketplace and platform economy terms to maximise retrieval comprehensiveness.

2.2.2. Screening

Search results from both databases were exported to the Parsifal platform (https://parsif.al/) and consolidated for the elimination of duplicates. Subsequently, two researchers independently conducted initial screening based on title and abstract using the pre-established inclusion and exclusion criteria. Disagreements between the two raters were resolved through discussion and, where necessary, through consultation with a third rater. Inter-rater agreement was measured using the Cohen's Kappa coefficient, with a value of $\kappa \geq 0.80$ set as the acceptable agreement threshold (Landis & Koch, 1977). Inter-rater agreement was high (Cohen's $\kappa = 0.86$), exceeding the pre-established threshold of 0.80 and indicating near-perfect agreement between the two independent raters. The inclusion and exclusion criteria used in the screening process are presented in full in Table 2.

2.2.3. Eligibility assessment

All articles that passed the initial screening stage were subsequently retrieved in full text and subjected to comprehensive full-text reading for final eligibility assessment. At this stage, articles were evaluated more deeply for their alignment with the research questions, including the relevance of the geographic context, the depth of discussion regarding e-commerce adoption, and the quality of the methodology employed. Articles that did not meet all inclusion criteria at the full-text reading stage were excluded,

with the reasons for exclusion documented systematically in accordance with the PRISMA 2020 guidelines (Page et al., 2021).

2.2.4. Quality appraisal

Methodological quality appraisal was conducted on all articles that passed the eligibility assessment to ensure the scientific integrity of the literature included in this review. Assessment was performed using the Mixed Methods Appraisal Tool (MMAT), which enables consistent evaluation of methodological quality across various types of research designs, including quantitative, qualitative, and mixed-method studies alike (Hong et al., 2018). Each article was evaluated on five primary dimensions: (1) clarity of research questions and appropriateness of design; (2) representativeness of samples or cases; (3) validity of variable measurement; (4) management of missing data; and (5) reliability of the analysis conducted. Articles scoring below the established minimum threshold were excluded from the review, while articles with moderate scores were retained but categorised within the recorded quality assessment. Methodological quality appraisal was conducted using the Mixed Methods Appraisal Tool (MMAT) version 2018 across seven criteria items (Q1–Q7): (Q1) Is there a clear research question? (Q2) Is the data collected appropriate to answer the research question? (Q3) Is the sampling strategy relevant to the research question? (Q4) Is the sample representative of the target population? (Q5) Are the measurements valid and reliable? (Q6) Is the risk of nonresponse bias low? and (Q7) Is the statistical analysis appropriate to the study design? Each criterion was scored dichotomously, where a response of 'Yes' was assigned a score of 1, while responses of 'No' or 'Can't Tell' were each assigned a score of 0, reflecting the inability to confirm methodological adequacy on that criterion. A minimum total score of 3 was set as the acceptable threshold for inclusion in the final synthesis, consistent with established MMAT scoring conventions (Hong et al., 2018). The MMAT quality appraisal scores for each of the 11 included studies are presented in Table 3. All 11 studies met the minimum quality threshold and were retained in the final synthesis. Scores ranged from satisfactory to strong across the seven MMAT criteria, with no studies excluded at this stage. The table below provides full transparency of each study's item-by-item scoring and total quality rating.

2.3. Data analysis: inductive thematic synthesis

Data analysis was conducted using the inductive thematic synthesis approach developed by Thomas and Harden (2008). This approach was selected for its capacity to generate deep and inclusive understanding from methodologically diverse literature encompassing quantitative, qualitative, and mixed-method studies which is commonly encountered in literature reviews in the fields of management and agribusiness (Flemming et al., 2019). Inductive thematic synthesis comprises three primary stages: (1) free line-by-line coding of findings from each included article; (2) development of descriptive themes from the grouping of codes sharing similar meanings; and (3) generation of analytical themes that reflect the deeper interpretation and meaning of the overall body of literature.

The coding and theme development process was conducted iteratively by two independent researchers using a standardised data extraction matrix encompassing the following dimensions: author and year,

Table 3. MMAT quality appraisal scores.

No.	Author	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Total score
1	Tian and Wu (2024)	1	1	1	0	0	1	1	5
2	Novanda et al. (2021)	1	1	1	0	1	0	1	5
3	Qiu et al. (2025)	1	1	1	1	0	0	1	5
4	Zhang and Jin (2020)	1	1	1	1	0	0	1	5
5	Yaochen et al. (2025)	1	1	1	0	1	0	1	5
6	Fagnani et al. (2023)	1	1	1	0	0	1	0	4
7	Chen et al. (2018)	1	1	1	0	1	0	1	5
8	Meyer et al. (2022)	1	1	1	0	0	0	1	4
9	Kabra and Dass (2024)	1	1	0	0	0	0	1	3
10	Kumar (2022)	1	1	1	0	1	0	1	5
11	Wen et al. (2024)	1	1	1	1	1	0	1	6

Source: Authors' own work.

country and context, study design, population and sample, theoretical framework employed, barriers identified, opportunities identified, and key findings. The resulting themes were subsequently verified and agreed upon through inter-researcher consensus discussions. Descriptive statistical analysis was employed to provide an overview of the distributional characteristics of the included literature, encompassing temporal, geographic, and methodological distributions.

3. Results

3.1. Study selection

At the initial selection stage, a number of studies were excluded based on the pre-established exclusion criteria. The literature search yielded 599 documents from two databases, namely Scopus (n = 397) and Web of Science (n = 202). Following the identification process, 163 duplicate records were removed, leaving 436 documents for the screening stage. At this stage, a number of documents were eliminated based on publication type: conference papers, proceedings papers, conference reviews, book chapters, reviews, books, notes, retracted articles, erratum, letters, and short surveys (n = 193); publication year outside the 2016–2026 window (n = 25); and non-English language (n = 18), leaving 200 documents eligible for title and abstract screening. Of these, 178 documents were excluded because their titles and abstracts were not relevant to the research focus. Following this screening process, 22 studies proceeded to the full-text retrieval stage.

The second selection stage was conducted by reading the full texts of the successfully retrieved studies and evaluating them against all established inclusion and exclusion criteria. Of the 22 studies for which full texts were sought, 10 studies could not be retrieved. Retrieval was attempted through multiple channels: (1) direct access via institutional subscriptions at Universitas Brawijaya; (2) interlibrary loan (ILL) requests submitted through the university library system; (3) searches on open-access repositories including ResearchGate, Academia.edu, and Unpaywall; and (4) direct email contact with corresponding authors requesting manuscript copies. Despite these efforts, the 10 studies remained inaccessible within the timeframe of this review, primarily due to restricted publisher access, non-responsive authors, or unavailability on open-access platforms. It is acknowledged that the inability to retrieve these studies may introduce a degree of selection bias, as potentially relevant findings contained within these inaccessible articles could not be incorporated into the synthesis. This limitation is noted as a constraint on the comprehensiveness of the review and is further discussed in Section 7 (Limitations).

Of the 12 remaining studies assessed for eligibility, 1 was excluded following full-text evaluation for not being related to e-commerce, leaving 11 studies for the final synthesis. At the eligibility stage, all remaining studies met the established inclusion criteria and were included. These 11 studies formed the basis for identifying and synthesising the various factors relevant to the research topic. The study selection process is presented in Figure 1.

3.2. Thematic synthesis of included studies

Based on systematic extraction from the 11 journal articles that met the inclusion criteria, five principal thematic clusters were derived through inductive thematic synthesis. These themes represent the dominant patterns in the literature concerning e-commerce adoption in livestock product marketing in developing countries. Table 4 presents the mapping of each included study along with the themes and theoretical frameworks employed. It should be noted that the review encompasses both supply-side adoption dynamics (e-commerce adoption by producers, MSMEs, and agribusiness actors) and broader market-level dimensions including consumer behaviour, food safety regulation, value-chain resilience, and ICT policy impacts. This broader scope reflects the multidimensional nature of e-commerce adoption in the livestock sector, where producer-level decisions are embedded within consumer markets, regulatory environments, and supply chain systems. Findings should therefore be interpreted with this dual supply-side and market-level scope in mind, rather than exclusively as evidence of producer-level e-commerce adoption behaviour.

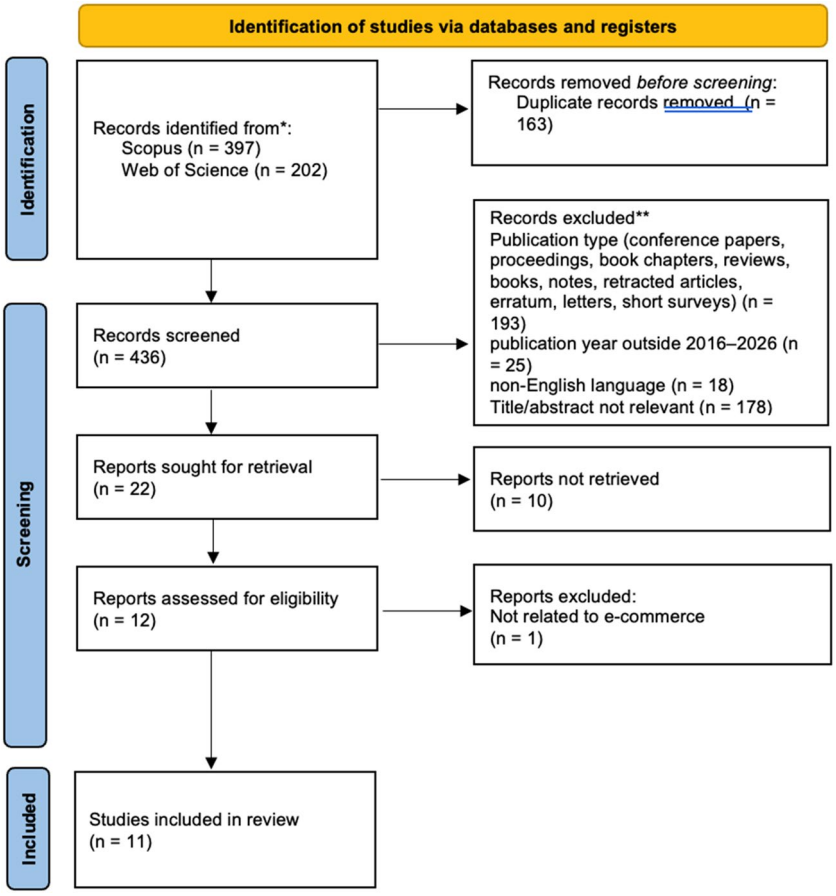


Figure 1. Prisma 2020 flow diagram.*Source:* Authors' own work.

3.3. Theme 1: E-commerce adoption and behavioural factors

Two studies in this review examined the antecedents and behavioural factors of e-commerce adoption among livestock sector actors, particularly smallholder farmers and micro, small, and medium enterprises (MSMEs).

Tian and Wu (2024) investigated 271 nomadic herders in Inner Mongolia, China, employing the Theory of Planned Behavior (TPB) to analyse how e-commerce participation mediates grassland utilisation pressure. The study findings indicate that the rate of e-commerce participation among herders remains relatively low, with only approximately 15% of respondents actively using online marketing channels. Reported barriers include limited mobile network signal coverage in remote grassland areas, limited access to capital, and the relatively small scale of operations.

Novanda et al. (2021) analysed the factors influencing e-commerce use among 31 dairy, meat, and egg MSMEs in Bengkulu, Indonesia, using the Technology-Organization-Environment (TOE) framework via PLS-SEM. The research findings indicate that external environmental factors particularly changes in consumer behaviour during the pandemic and competitive market pressure together with organisational readiness, played a significant role in driving e-commerce adoption. Conversely, technological factors were not found to be statistically significant within the research model.

3.4. Theme 2: consumer behaviour and market access

Three studies examined how e-commerce influences consumer behaviour and expands market access for livestock products. Qiu et al. (2025) employed a binary probit model with mediation analysis on 2,805 rural farming households in China to examine the relationship between e-commerce participation and

Table 4. Thematic synthesis of studies.

No.	Author	Title	Country	Theme	Theoretical framework
1	Tian and Wu (2024)	E-Commerce Participation, Subjective Norms and Grassland Utilization Pressure: An Empirical Evidence of Herdsmen in Inner Mongolia, China	China	E-Commerce Adoption & Behavioral Factors	Theory of Planned Behavior (TPB)
2	Novanda et al. (2021)	Investigation and Analysis on the Factors Influencing the Use of E-Commerce SUSDAGTEL SMEs in Bengkulu, Indonesia	Indonesia	E-Commerce Adoption & Behavioral Factors	Technology-Organization-Environment (TOE) Framework
3	Qiu et al. (2025)	E-commerce and green food consumption of rural residents: implications for sustainable rural food systems	China	Consumer Behavior & Market Access	Information Asymmetry Theory, Cognitive Coordination Theory, Baron & Kenny Mediation Framework
4	Zhang and Jin (2020)	Hedonic valuation of country of origin in the Chinese dairy market	China	Consumer Behavior & Market Access	Hedonic Price Theory, Consumer Preference Theory
5	Yaochen et al. (2025)	Omnichannel and Product Quality Attributes in Food E-Retail: A Choice Experiment on Consumer Purchases of Australian Beef in China	China	Consumer Behavior & Market Access	Lancastrian Consumer Theory, Random Utility Theory, Signalling Theory
6	Fagnani et al. (2023)	A novel approach in public health surveillance: searching the illegal dairy trade in Facebook	Brazil	Digital Surveillance & Food Safety Regulations	Internet-based biosurveillance
7	Chen et al. (2018)	Assessment of Dairy Product Safety Supervision in Sales Link: A Fuzzy-ANP Comprehensive Evaluation Method	China	Digital Surveillance & Food Safety Regulations	Information Asymmetry Theory, Moral Hazard Theory, Food Safety Governance Framework
8	Meyer et al. (2022)	Hotspots of vulnerability and disruption in food value chains during COVID-19 in South Africa	South Africa	Value Chain Resilience & Digital Pivoting	Value Chain Analysis, Agrifood System Resilience Framework, Organizational Pivoting Theory
9	Kabra and Dass (2024)	Licious's Response to the COVID-19 Pandemic: A Step Towards Crisis Communication?	India	Value Chain Resilience & Digital Pivoting	Crisis and Emergency Risk Communication (CERC) Model
10	Kumar (2022)	ICT Effect on Supply Chain Performance in the Dairy Industry: A Study in the Indian Context	India	ICT & Supply Chain Integration	SCM and ICT performance linkage framework
11	Wen et al. (2024)	Revitalizing Agricultural Economy Through Rural E-Commerce? Experience from China's Revolutionary Old Areas	China	ICT & Supply Chain Integration	Classical Political Economy, Agricultural Industrialization Theory, Information Asymmetry Framework

Source: Authors' own work.

green food consumption. The study identified three primary mechanisms: increases in household income, improved access to information, and improvements in logistical infrastructure.

Zhang and Jin (2020) employed a hedonic price model to analyse 797 UHT dairy products and 636 infant formula products from several B2C e-commerce platforms in China. The research findings indicate that product attributes such as country of origin, organic labels, and protein content influence price premiums for dairy products on e-commerce platforms.

Yaochen et al. (2025) employed a discrete choice experiment on 872 urban consumers in China to analyse consumer preferences for Australian beef in an omnichannel retail context. The study findings indicate that consumers demonstrated preferences for a retail system that integrates digital platforms with physical stores, alongside product attributes such as traceability and product origin certification.

3.5. Theme 3: digital surveillance and food safety regulations

Fagnani et al. (2023) analysed informal dairy trade conducted through Facebook Marketplace in Brazil, employing an internet-based biosurveillance approach. The study demonstrated that social media platforms may function as distribution channels for informal dairy products that operate beyond the monitoring capacity of conventional surveillance systems.

Chen et al. (2018) evaluated the safety supervision of dairy products across 1,501 online and offline businesses in China, utilising a Fuzzy Analytic Network Process (Fuzzy-ANP) model. The study revealed a range of systemic weaknesses in the online sales supervision framework, including limited verification of seller qualifications and substantial cross-regional regulatory challenges.

3.6. Theme 4: value chain resilience and digital pivoting

Two studies examined how digitalisation and e-commerce contributed to value chain resilience and adaptive capacity during periods of disruption, particularly during the COVID-19 pandemic. Meyer et al. (2022) analysed hotspots of vulnerability and disruption in South African food value chains during the COVID-19 pandemic using a food system resilience framework. The study found that actors with stronger digital integration and institutional coordination were better able to adapt to market disruptions, mobility restrictions, and logistical bottlenecks. The emergence of online livestock auctions and digitally coordinated supply systems demonstrated the role of e-commerce as a mechanism for maintaining market continuity during crises.

Kabra and Dass (2024) examined the response of Licious, an Indian meat e-commerce company, during the COVID-19 pandemic through the lens of crisis communication and organisational adaptation. The study showed that the integration of digital ordering systems, cold chain logistics, and strategic communication enabled the company to respond effectively to rapid increases in consumer demand during lockdown periods. The findings highlight the importance of digital infrastructure and supply chain flexibility in strengthening resilience within livestock product marketing systems.

3.7. Theme 5: ICT and supply chain integration

Kumar (2022) examined the relationship between the use of information and communication technology (ICT) and supply chain performance in the dairy industry in India using multiple regression analysis. The research findings indicate that ICT use is associated with improvements in logistical efficiency, delivery punctuality, and procurement efficiency.

Wen et al. (2024) employed a difference-in-differences approach across 1,230 districts in China to evaluate the impact of rural e-commerce policies on the agricultural economy. The research findings indicate that the policy increased agricultural sector value added and drove productivity improvements in rural areas.

4. Discussion

4.1. E-Commerce adoption among livestock sector actors: between situational pressure and structural readiness

The findings from the two studies constituting Theme 1 collectively portray a landscape of e-commerce adoption in the livestock sector of developing countries that remains characterised by structural unevenness. Tian and Wu (2024) found that only 15% of nomadic herders in Inner Mongolia actively use digital marketing channels, while Novanda et al. (2021) identified that organisational readiness and external environmental pressure rather than technological capacity constitute the primary drivers of e-commerce adoption among livestock MSMEs in Indonesia. Both findings collectively indicate that barriers to e-commerce adoption in the livestock sector are multidimensional and cannot be reduced solely to the question of technology availability.

This result is consistent with the fundamental premise of the Technology-Organization-Environment (TOE) framework, which affirms that technology adoption is a function of three interacting sets of factors: organisational technological capacity, internal readiness, and external environmental pressure (Tornatzky & Fleischer, 1990). These findings are consistent with the seminal arguments of Zeng et al. (2017) that the barriers to e-commerce adoption in the livestock sector are multidimensional, cannot be reduced only to the availability of technological infrastructure. The fact that technological factors were non-significant in Novanda et al. (2021) study, while environmental and organisational factors were dominant, suggests that in the context of livestock MSMEs in developing countries, the increasingly available digital infrastructure (e.g. smartphone penetration) does not automatically drive e-commerce adoption unless accompanied by adequate managerial capacity readiness and sufficient market incentives. This finding shifts the policy implication focus from mere digital infrastructure investment towards strengthening the organisational capacity of small-scale livestock business actors.

On the other hand, the use of the Theory of Planned Behavior (TPB) by Tian and Wu (2024) offers a different yet complementary perspective. The finding that social norms and government norms moderate the relationship between e-commerce and herder behaviour affirms the relevance of the socio-cultural dimension in the technology adoption process within traditional livestock communities. This is consistent with the arguments advanced in the innovation diffusion literature, which holds that the adoption of new technologies in agrarian communities is strongly influenced by social networks and the social legitimacy embedded in the innovation (Rogers, 2003). Consequently, e-commerce adoption assistance programmes designed without consideration of local community social structures and norms tend to encounter significant resistance, even when the technical infrastructure is already in place.

The broader implication of both findings is the need for a contextual and phased e-commerce adoption approach: beginning with the strengthening of digital literacy and managerial capacity, proceeding to the construction of a social trust ecosystem towards digital platforms, and subsequently advancing to the development of e-commerce business models appropriate to the scale and characteristics of livestock enterprises in each respective context. The e-commerce demonstration county programme proven effective in China as reported by Tian and Wu (2024) may serve as a relevant reference model to be adapted in other developing countries, including Indonesia.

4.2. Consumer behaviour and market access dynamics: implications for digital marketing strategies for livestock products

The three studies constituting Theme 2 consistently demonstrate that e-commerce functions not only as a distribution channel, but also as a mechanism for value formation and shaping consumer preferences towards livestock products. Qiu et al. (2025) demonstrated that e-commerce participation increases green food consumption through three mediating pathways: income increases, strengthened access to information, and improvements in logistical infrastructure. This finding is in line with the empirical evidence of Yu et al. (2023) which shows that the application of e-commerce allows small-scale egg farmers to obtain premium prices through digital quality signaling mechanisms.

The contribution of Zhang and Jin (2020) through hedonic price analysis provides solid quantitative evidence regarding the structure of attribute values for dairy products on e-commerce platforms. The finding that organic labels and protein content significantly increase price premiums confirms that consumers shopping for livestock products online possess a higher willingness to pay for quality attributes that can be verified digitally. This directly supports the argument that the information asymmetry that has long been a primary barrier in traditional livestock product marketing can be effectively reduced through digital quality signalling mechanisms such as certification, labels, and traceability. The paradoxical finding regarding the low price premium for UHT dairy products from Europe simultaneously offers an important caveat: the reputation of country of origin is not universal, but is highly dependent on the product category and the historical context of consumer trust.

The study by Yaochen et al. (2025) enriches this understanding by demonstrating that urban consumers in developing countries experiencing middle-class growth seek not only online shopping convenience but also a seamless integration of digital and physical shopping experiences. The strong preference for omnichannel models with the highest willingness to pay for traceability labels indicates that the future of premium livestock product marketing in developing countries lies in the ability to build a digital trust ecosystem linking product origin information (farm-to-table transparency) with multi-channel shopping convenience. This is consistent with the argument of Signalling Theory that brands and certifications function as quality signals that reduce perceived consumer risk in transactions involving credence attributes attributes that cannot be assessed even after consumption.

In aggregate, the findings from Theme 2 provide clear strategic implications for farmers and business actors in this sector: investment in value-added attributes communicable digitally such as organic certification, QR code-based traceability systems, and verified origin labelling constitutes a prerequisite for deriving maximum economic benefit from e-commerce adoption. This simultaneously affirms that e-commerce adoption unaccompanied by the development of production capacity for high value-added products will yield only marginal benefits for smallholder farmers in developing countries.

4.3. The duality of E-commerce in food safety: digital surveillance opportunities and regulatory risks

The findings from Theme 3 present a perspective often overlooked in discussions of e-commerce adoption in the food sector: the dimension of food safety and regulatory governance. Fagnani et al. (2023) empirically demonstrated that the expansion of e-commerce particularly through social media can unexpectedly extend the reach of trade in livestock products that fail to meet food safety standards. The finding that the highest illegal dairy trade activity was found in areas with better technological access reveals a digitalisation paradox: the ease of technological access intended to enhance market efficiency simultaneously lowers the barrier to entry for unregistered informal traders.

This paradox must be understood within the framework of Information Asymmetry Theory: in an unregulated digital environment, consumers face greater information asymmetry than in traditional markets, due to their inability to verify the authenticity of labels and certifications on livestock products sold online. Chen et al. (2018) confirmed this problem from a regulatory perspective, finding that regional economic development level rather than platform type or business type is the primary determinant of online dairy product supervision quality. This finding implies that the regulatory capacity gap arising from unequal development between regions will become more exposed and potentially worsen the quality disparity of livestock products circulating in digital markets.

On the positive side, the internet-based biosurveillance approach demonstrated by Fagnani et al. (2023) opens a new perspective on how digital technology can be utilised to strengthen rather than merely weaken food safety surveillance systems. The potential for replicating social media data-based biosurveillance models in other developing countries is highly significant, given the considerably lower operational costs compared to conventional inspections and the monitoring coverage that can encompass entire national territories simultaneously. The policy implication arising from both studies is the urgency of developing a regulatory framework for e-commerce of livestock products that not only focuses on facilitating trade, but also proactively integrates digital surveillance mechanisms adaptive to the characteristics of online trade.

4.4. Value chain resilience and digital adaptation capacity: lessons from crisis

The two studies in Theme 4 provide valuable lessons regarding the role of e-commerce and digitalisation as resilience mechanisms in livestock product value chains when confronted with large-scale external disruptions. Meyer et al. (2022) demonstrated that agri-food value chains in South Africa exhibited varying adaptive capacities depending on the level of digitalisation and institutional coordination capacity possessed prior to the onset of the crisis. The case of online livestock auctions adopted in response to mobility restrictions during the COVID-19 pandemic illustrates how digital pivoting can occur rapidly when its supporting ecosystem digital infrastructure, inter-actor trust, and industry coordination has been previously established.

The Licious case in India, examined by Kabra and Dass (2024), complements this perspective from the standpoint of an established e-commerce company. The demand surge of up to 300% during lockdown successfully managed through the integration of cold chain, third-party logistics, and aggressive crisis communication strategies demonstrates that the resilience of livestock product e-commerce businesses in developing countries is highly determined by the depth of technology integration throughout the value chain, from raw material procurement, processing, cold storage, last-mile distribution, to communication with end consumers. This capacity cannot be built instantaneously in crisis conditions, but is rather an accumulation of long-term investment in supply chain digital capabilities.

Comparatively, both studies underline a fundamental distinction between reactive digital pivoting conducted as a crisis response without adequate digital foundations and digital resilience built proactively through sustained investment in e-commerce capabilities. This lesson is particularly relevant for developing countries whose livestock sectors remain in the early stages of digital transformation: crises such as the COVID-19 pandemic can function as accelerators of e-commerce adoption, but only for actors who already possess a sufficient foundation of digital capabilities. For those who do not, crises can permanently worsen their competitive position.

4.5. ICT as supply chain performance infrastructure: policy implications for developing countries

The two studies constituting Theme 5 provide strong empirical evidence regarding the positive impact of ICT investment and e-commerce policies on supply chain performance and aggregate value added in the livestock sector. Kumar (2022) documented a significant positive relationship between ICT adoption encompassing ERP, RFID, GPS, and Smart Milk Procurement Systems and all measured supply chain performance indicators in the dairy industry. Meanwhile, Wen et al. (2024) produced causal evidence through a quasi-experimental design that rural e-commerce policies implemented by the Chinese government were capable of increasing agricultural sector value added by approximately 3%, driving livestock output growth, and improving total factor productivity in rural areas.

Both findings collectively indicate that the impact of e-commerce on the livestock sector is not merely microeconomic in terms of increasing individual farmer income but also has significant macroeconomic dimensions through improvements in system efficiency and aggregate value added. The transmission mechanisms identified in the Wen et al. (2024) study namely the expansion of digital market access and reduction of transaction costs are consistent with the predictions of Information Asymmetry Theory that market efficiency will improve when information barriers between producers and consumers can be systematically eliminated. In the context of the livestock sector in developing countries, where inefficiencies in traditional marketing chains frequently result in large price disparities between the farmer level and end consumers, the potential impact of e-commerce on the efficiency of value distribution is highly substantive.

The limitations of Kumar's (2022) study covering only large-scale dairy plants in certain regions restrict the generalisation of these findings to the context of small and informal farmers. Nevertheless, the trajectory demonstrated by technologies such as the Smart Milk Procurement System which successfully integrated smallholder farmers into the digital supply chain through real-time milk quality measurement and digital payment systems provides a prototype that can be adapted for livestock agribusiness contexts in other developing countries. The most fundamental policy implication of both studies is the need for integrated rural e-commerce policies, rather than mere partial digitalisation programmes, to produce transformative and significant impact for the livestock agribusiness ecosystem.

5. Theoretical gaps and future research directions

This systematic review identifies several significant theoretical and methodological gaps in the literature on e-commerce adoption in the livestock sector of developing countries, which simultaneously open important avenues for a future research agenda.

First, the absence of mainstream technology adoption frameworks in the majority of included studies constitutes the most conspicuous theoretical gap. None of the included studies employed the Technology Acceptance Model (TAM), the Unified Theory of Acceptance and Use of Technology (UTAUT), or the Diffusion of Innovation (DOI) the three most dominant frameworks in the general e-commerce adoption literature. Only two studies employed technology adoption frameworks explicitly (TPB and TOE), while the remainder predominantly used frameworks from agricultural economics, supply chain management, or consumer marketing disciplines. This absence is not incidental but reflects deeper structural characteristics of the field. Three explanations are proposed: (1) the disciplinary dominance of agricultural economics, wherein the primary analytical lens centres on market efficiency and value chain dynamics rather than individual-level technology acceptance behaviour; (2) the poor fit between existing adoption models and the lived realities of smallholder livestock farmers TAM and UTAUT assume that users have basic digital skills, reliable internet access, and make technology decisions independently, yet smallholder farmers in developing countries typically face capital shortages, depend on community-based decision-making, and operate within value chains still controlled by informal intermediaries, conditions under which these models lose much of their explanatory power, as confirmed by Novanda et al. (2021), who found that technological factors were statistically non-significant in their TOE-based analysis; and (3) the early developmental stage of this research field, where scholars have yet to move beyond their home disciplines agricultural economists and food systems researchers naturally draw on familiar theoretical tools before cross-disciplinary borrowing from information systems becomes standard practice.

Second, the dominance of the Chinese context in the identified literature (7 out of 11 studies) creates a serious geographic gap that warrants critical reflection on the transferability of the review's findings. While China's agricultural e-commerce experience is highly instructive, its findings cannot be straightforwardly transferred to other developing country contexts due to three structural differences. First, in terms of digital infrastructure, China possesses near-universal mobile internet penetration, an extensive cold chain logistics network, and government-subsidised last-mile delivery systems conditions that stand in stark contrast to Sub-Saharan Africa, where rural internet penetration remains critically low and cold chain infrastructure is largely absent, and to Southeast Asia and Latin America, where pronounced urban-rural digital divides continue to constrain platform-based e-commerce adoption. Second, regarding government policy, China's rural e-commerce transformation has been driven by large-scale, coordinated state intervention including the e-commerce demonstration county programme documented by Wen et al. (2024) a level of policy coordination and public investment that is largely absent in most African and Latin American countries, where agricultural digitalisation policies remain fragmented and primarily donor-driven. Third, concerning livestock sector structure, Chinese smallholder farming operates within relatively well-integrated cooperative systems that facilitate collective e-commerce adoption, whereas livestock systems in Sub-Saharan Africa are predominantly characterised by extensive pastoralism, informal value chains, and weak institutional coordination conditions under which platform-based digital marketing models developed in the Chinese context are unlikely to be directly replicable. Future research must therefore explicitly target these underrepresented regions to generate findings that are genuinely generalisable to the broader developing country population.

Third, the existing literature employs almost exclusively cross-sectional designs, which do not permit the capture of temporal dynamics in the e-commerce adoption process. Given that e-commerce adoption in the livestock sector is a process that unfolds gradually and is influenced by the accumulation of experience, infrastructure development, and policy changes, longitudinal studies are critically needed to understand the trajectory and the factors that determine long-term adoption success.

Fourth, the perspective of smallholder farmers as primary subjects rather than merely as policy objects is severely underrepresented in the literature. Indeed, of the eleven studies synthesised in this review, only two engaged directly with smallholder or MSME-level actors as primary units of analysis, while the remainder examined consumer markets, established companies, or macro-level policy effects; this imbalance means that the review's conclusions regarding 'opportunities for smallholder farmers' are necessarily inferential rather than directly evidenced. The majority of studies examining the supply side focus on medium-to-large scale companies or cooperatives. Future research needs to more extensively explore the experiences, barriers, and adaptive strategies of smallholder farmers who are directly involved in, or are considering, e-commerce adoption, including exploration of collective e-commerce models based on farmer groups that may be more appropriate to the small-scale business context.

Fifth, none of the studies in this review explicitly examined the gender and social inclusion dimensions of e-commerce adoption in livestock products. Given the highly significant role of women in the livestock value chain in many developing countries ranging from the care of small livestock, dairy product processing, to local-level marketing research that integrates a gender perspective would make an important contribution to the design of inclusive e-commerce adoption programmes. Gender, youth, and sustainability dimensions are particularly important in shaping e-commerce adoption in livestock product marketing because they determine who actually benefits from digitalisation. Women in developing-country livestock systems frequently face compounding disadvantages: restricted access to smartphones and mobile internet, limited control over household income generated from livestock sales, lower digital literacy relative to male household members, constrained mobility that limits participation in physical markets and digital training programmes, and reduced decision-making power in adoption of new technologies and business models. These structural disadvantages mean that e-commerce programmes designed without gender-sensitive considerations risk reinforcing existing inequalities rather than reducing them. Youth, on the other hand, represent a critical enabling group: younger household members are more likely to possess digital skills, familiarity with online platforms, and willingness to experiment with new marketing channels, yet targeted programmes that leverage youth as digital intermediaries for smallholder households remain largely absent from the policy landscape. In relation to sustainability, e-commerce can function as an instrument for incentivising more environmentally responsible livestock practices by enabling

premium pricing for green, organic, and traceability-certified products, thereby creating a market-based pathway to sustainable production. Future research should explicitly integrate gender, youth, and environmental sustainability as analytical dimensions in studies of e-commerce adoption in livestock systems.

Sixth, the integration between environmental sustainability dimensions and e-commerce adoption in the livestock sector also constitutes a relatively unexplored area. The finding of Tian and Wu (2024) regarding the negative relationship between e-commerce and grassland utilisation pressure represents an interesting starting point, yet needs further development in a broader context, including the potential of e-commerce as an instrument for promoting more environmentally sound livestock practices through price premium mechanisms for green and organically labelled products.

6. Implications

The findings from this systematic review carry several significant policy implications for governments, development institutions, and industry actors in developing countries. First, policy interventions to promote e-commerce adoption in the livestock sector must be holistic and integrated, encompassing digital infrastructure strengthening (connectivity, cold chain, last-mile logistics), managerial capacity and digital literacy development for business actors, and the construction of a regulatory framework adaptive to the characteristics of digital trade in livestock products. Approaches focusing on only one dimension infrastructure alone, for instance have proven insufficient to generate meaningful e-commerce adoption.

Second, governments in developing countries need to develop digital food safety surveillance systems capable of keeping pace with the rapid development of online livestock product trade. The experiences of Brazil and China, as presented in this review, demonstrate that the delayed regulatory adaptation to the digital ecosystem can create serious food safety risks, both for consumers and for the overall reputation of domestic livestock products.

Third, government-designed agricultural e-commerce development programmes need to explicitly consider the need for differentiation of livestock products through value-added attributes communicable digitally. Facilitating organic certification, developing technology-based traceability systems, and building collective brands for local livestock products constitute institutional investments that will substantially enhance the competitiveness of livestock products from developing countries on both domestic and international e-commerce platforms. Fourth, financial inclusion remains a critical enabling condition for smallholder participation in digital livestock markets. Green banking innovation and inclusive agricultural finance can improve farmers' capacity to invest in market-oriented technologies, sustainable production systems, and digital business models (Siankwilimba et al., 2025). Policy frameworks for e-commerce promotion in the livestock sector must therefore be accompanied by accessible financing instruments, including agricultural credit schemes, digital payment infrastructure, and public-private investment programmes that lower the capital barriers confronting smallholder livestock enterprises.

7. Limitations

This systematic review has several limitations that must be acknowledged. First, the database coverage used is limited to Scopus and Web of Science, which may not fully represent all relevant literature, particularly research published in national journals of developing countries outside Scopus indexation. Future research may expand the search scope by integrating additional databases such as Google Scholar and regional databases. Second, the restriction to English-language articles may result in the omission of important contributions from local-language academic literature, particularly from countries such as China, Brazil, and Indonesia, which possess active and productive agricultural research communities in their respective national languages. Third, the majority of included studies employed cross-sectional designs with limited geographic coverage, such that the generalisation of findings to the overall developing country population should be undertaken with caution. Fourth, the literature included in this review reflects the state of research up to the mid-2020s, such that the very dynamic developments in the e-commerce ecosystem of the livestock sector particularly in the post-COVID-19 pandemic period may not yet be fully reflected in the resulting synthesis. Fifth, although smallholder farmers are foregrounded in the title and keywords of this review, the underlying evidence base predominantly derives

from consumer-level studies, medium-to-large agribusiness enterprises, dairy processing companies, food safety regulatory analyses, and ICT policy evaluations, rather than direct empirical engagement with smallholder farmers themselves. Only two of the eleven included studies (Novanda et al., 2021; Tian & Wu, 2024) directly examined smallholder or MSME-level adoption behaviour. Consequently, smallholder-specific evidence within this review remains thin, and readers should avoid interpreting the synthesised findings as uniformly representative of smallholder realities. The challenges and opportunities identified should instead be understood as drawn from a broader livestock e-commerce ecosystem encompassing producers, consumers, intermediaries, and regulators, within which smallholder farmers constitute only one, and currently underrepresented, category of actor.

Acknowledgements

This work was supported by Lembaga Pengelola Dana Pendidikan (LPDP) Scholarship From Kementerian Keuangan Republik Indonesia (Kemenkeu RI). The authors are grateful for the facilities from Universitas Brawijaya.

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CRediT: **Fahmi Alif Ardiansyah**: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft; **Ahmad Ridwan**: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Writing – review & editing; **Siti Azizah**: Conceptualization, Funding acquisition, Methodology, Project administration, Supervision, Validation, Writing – review & editing.

Disclosure statement

No potential conflict of interest was reported by the authors.

Funding

This research was supported by the Indonesia Endowment Fund for Education (Lembaga Pengelola Dana Pendidikan/ LPDP), Ministry of Finance of the Republic of Indonesia, and Universitas Brawijaya.

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Data availability statement

Data sharing is not applicable to this article as no data were created or analysed in this study.

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